1) Water Jug Problem. (4 gallon and 3 gallon Jugs)

Goal: Get exactly 2 gallons of water in the 4-gallon jug.

Soln steps:

Step	Action	4 gallon jug	3 gallon jug
1	Fill the 3 gallon jug	0	3
2	Pour into the 4 gallon jug	3	0
3	Fill the 3 gallon jug again	3	3
4	Pour into the 4 gallon jug until full	4	2
5	Empty the 4 gallon jug	0	2
6	Pour the remaining water into the 4-gallon jug	2	0

Result: Exactly 2 gallons are obtained.

2 Mars Rover Problem

i) Percepts: The rover receives info through camera sensors, GPS-like navigation systems and scientific intruments. These detect environmental conditions

ii) Operating Enviroment: Mars is a partially observable uncertain, dynamic, sequential and mostly continous enviroment. Communication with Earth is delayed

iii) Actions: The rover can move, turn, avoid obstacles, collect soil and rock samples, take photographs and analyse samples.

iv) Performance Evaluation: Performance is measured by successful navigation, useful scientific discoveries, accurate sample analysis energy efficiency, and mission completion

v) Suitable Agent Architecture: A model based, goal based agent is most suitable. It maintains an internal map of its surroundings and plans an internal map of obstacles to avoid and conserve energy.

3) 8- Queens Problem

Goal: Place 8 queens on an 8 x 8 chessboard so that no two queens attack each other

Problem Components :

1) Initial State : An empty chessboard

2) State Space: All possible arrangements of 8 queens on the board

3) Actions: Place one queen in a safe square, preferably one queen per column

4) Transition Model: Adds a queen to the selected square.

5) Goal test: All 8 queens are placed, with no two sharing a row, column, or diagonal.

6) Path Cost: Each queen placement has a cost of 1; the total cost is 8 placements

4) OLA Cab Booking Problem:

Suitable Agent: A goal based problem solving agent is suitable because it selects a cab and route to reach the customers destination according to their preferences

Pseudo code

```
START
  Input source and destination
  Input cab preference and budget
  Find available cabs
  Calculate fare and travel time
  Remove cabs outside the budget
  Select the best cab based on Preference
  Display cab type, fare and arrival time
  if customer confirms:
      Book the cab
  else:
      Select another cab
  STOP
```

Explanation: The agent considers the destination, cabs, fare before selecting a suitable fare. Booking occurs only after customer confirmation.

5 Uniform Cost Search (UCS) for the Ddivery Network

UCS expends the node with the lowest cumulative path cost g(n) it is suitable because the network has different route and the objective is to minimize the total cost.

From the weighted graph

- S -> A = 1
- S -> G = 12
- A -> B = 3
- A -> C = 1
- B -> D = 3
- C -> D = 1
- C -> G = 2
- D -> G = 3

The least - cost path is

S -> A -> C -> G

Total cost

$1 + 1 + 2 = \boxed{4}$

Interpretation:

The minimum delivery cost is 4 cost units. The route is cheaper than the direct route from s to h, which costs 12 units. UCS guarantees an optimal solution when all edge costs are non-negative.